

Spring 2006

Problem 1 (Sp06). Let G be the subgroup of the free abelian group $\mathbb{Z}^4$ consisting of all integer vectors (x, y, z, w) such that $2x + 3y + 5z + 7w = 0$.

1. Determine a linearly independent subset of G which generates G as an abelian group.
2. Show that $\mathbb{Z}^4/G$ is a free abelian group and determine its rank.

Problem 2 (Sp06). Find all real numbers c such that the initial value problem

$$y'' - cy' + 16y = 0 \quad y(0) = y(1) = 1$$

has no solution.

Problem 3 (Sp06). Let $S = \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_1 + \dots + x_n = 0\}$. Find the $n \times n$ matrix P of the orthogonal projection from $\mathbb{R}^n$ onto S .

Problem 4 (Sp06). Find all holomorphic functions $f: \mathbb{D} \rightarrow \mathbb{C}$ such that $f\left(\frac{1}{n} + \frac{i}{e^n}\right)$ is real for all integers $n \geq 2$.

Problem 5 (Sp06). Consider the following four commutative rings:

$$\mathbb{Z} \quad \mathbb{Z}[x] \quad \mathbb{R}[x] \quad \mathbb{R}[x, y]$$

Which of these rings contains a nonzero prime ideal that is not a maximal ideal?

Problem 6 (Sp06). Let $u: \mathbb{R} \rightarrow \mathbb{R}$ be a function for which there exists $B > 0$ such that

$$\sum_{k=1}^{N-1} |u(x_{k+1}) - u(x_k)|^2 \leq B$$

for all finite increasing sequences $x_1 < x_2 < \dots < x_N$. Show that u has at most countably many discontinuities.

Problem 7 (Sp06). Suppose that $A \in \mathrm{SL}_2(\mathbb{R})$ does not have a real eigenvalue. Show that there exists $B \in \mathrm{SL}_2(\mathbb{R})$ such that BAB^{-1} equals a rotation matrix $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ for some $\theta \in \mathbb{R}$.

Problem 8 (Sp06). Let $f: \mathbb{D} \rightarrow \mathbb{C}$ be holomorphic, and suppose that the restriction of f to $\mathbb{D} - \{0\}$ is injective. Prove that f is injective.

Problem 9 (Sp06). Let p be a prime and G a finite noncyclic group of order p^m for some m . Prove that G has at least $p + 3$ subgroups.

Problem 10 (Sp06, Fa10, Sp13, Sp21). Prove that the intersection of a decreasing sequence of nonempty closed connected subsets of a compact metric space is nonempty and connected. Give an example to show that this is false if the assumption that the space is compact is dropped.

Problem 11 (Sp06). Let $\mathbb{F}_2$ be the field of 2 elements and n a prime. Show that there are exactly $(2^n - 2)/n$ degree- n irreducible polynomials in $\mathbb{F}_2[x]$.

Problem 12 (Sp06, Sp25). Evaluate the integral $\int_{-\infty}^{\infty} \frac{e^{itx}}{e^x + e^{-x}} dx$ for $t > 0$.

Problem 13 (Sp06). Let n be a positive integer, and let $GL_n(\mathbb{R})$ be the group of invertible $n \times n$ matrices. Let S be the set of $A \in GL_n(\mathbb{R})$ such that $A - I$ has rank ≤ 2 . Prove that S generates $GL_n(\mathbb{R})$ as a group.

Problem 14 (Fa82, Sp06). 1. Prove that there is no continuous map from the closed interval $[0, 1]$ onto the open interval $(0, 1)$.

2. Find a continuous surjective map from the open interval $(0, 1)$ onto the closed interval $[0, 1]$.

3. Prove that no map in Part 2 can be bijective.

Problem 15 (Sp06). Let A be the subring of $\mathbb{R}[x]$ consisting of polynomials $f(x)$ such that $f'(0) = 0$. Is A a principal ideal domain?

Problem 16 (Sp06). Let m be a fixed positive integer.

1. Show that if an entire function $f: \mathbb{C} \rightarrow \mathbb{C}$ satisfies $|f(z)| \leq e^{|z|}$ for all $z \in \mathbb{C}$, then

$$|f^{(m)}(0)| \leq \frac{m!e^m}{m^m}$$

2. Prove that there exists an entire function f such that $|f(z)| \leq e^{|z|}$ for all z and

$$|f^{(m)}(0)| = \frac{m!e^m}{m^m}$$

Problem 17 (Sp06). Let $\langle \cdot, \cdot \rangle$ be the standard Hermitian inner product on $\mathbb{C}^n$. Let A be an $n \times n$ matrix with complex entries. Suppose $\langle x, Ax \rangle$ is real for all $x \in \mathbb{C}^n$. Prove that A is Hermitian.

Problem 18 (Sp06). Find a bounded nonconvergent sequence of real numbers (a_n) such that

$$|2a_n - a_{n-1} - a_{n+1}| \leq \frac{1}{n^2}$$

for all $n \geq 2$.